

DDA3020 Machine Learning

Exercise: Probability and Linear Algebra

April 2026

Instructions

This exercise covers the main ideas of Probability and Linear Algebra, including independent events, random variables, entropy, Kullback-Leibler divergence, matrix inversion, and solving over/under-determined systems of linear equations. Show your working clearly for all calculation and derivation questions.

Useful Formulas

- Bayes' rule: $P(A | B) = \frac{P(B|A)P(A)}{P(B)}$
- Variance: $\text{Var}(X) = E(X^2) - (E(X))^2$
- Entropy: $H_P(\mathcal{X}) = -\sum_{x \in \mathcal{X}} P(x) \log P(x)$
- Matrix inverse: $\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \text{adj}(\mathbf{A})$
- Left-inverse (over-determined system $\mathbf{X}\mathbf{w} = \mathbf{y}$): $\mathbf{X}^\dagger = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$
- Right-inverse (under-determined system $\mathbf{X}\mathbf{w} = \mathbf{y}$): $\mathbf{X}^\dagger = \mathbf{X}^\top (\mathbf{X}\mathbf{X}^\top)^{-1}$

Questions

- Q1.** Let A and B be two independent events with $P(A) = 0.4$ and $P(B) = 0.5$. What is $P(A \cup B)$?
- A) 0.9
 - B) 0.7
 - C) 0.2
 - D) 0.8
- Q2.** If a random variable X follows the binomial distribution $\text{Bin}(N, \mu)$, what is its variance?
- A) $N\mu$
 - B) $N\mu(1 - \mu)$
 - C) $\mu(1 - \mu)$
 - D) $N(1 - \mu)$

- Q3.** Which of the following is true about Kullback-Leibler (KL) divergence $D_{P,Q}(\mathcal{X})$ between distributions P and Q ?
- A) It is a symmetric measure, i.e., $D_{P,Q}(\mathcal{X}) = D_{Q,P}(\mathcal{X})$.
 - B) $D_{P,Q}(\mathcal{X}) \leq 0$.
 - C) $D_{P,Q}(\mathcal{X}) = H_{P,Q}(\mathcal{X}) - H_P(\mathcal{X})$.
 - D) It represents the expected value of information from a source.
- Q4.** For a continuous random variable X , which of the following is true regarding its probability density function $p_X(x)$ and cumulative distribution function $F_X(x)$?
- A) $p_X(x) \leq 1$ for all x
 - B) $P(X = x) = p_X(x)$
 - C) $F_X(x) = \int_{-\infty}^x p_X(t)dt$
 - D) $\int_{-\infty}^{\infty} F_X(x)dx = 1$
- Q5.** Which of the following statements is equivalent to a square matrix A being nonsingular?
- A) The columns of A are linearly dependent.
 - B) $A\mathbf{x} = \mathbf{0}$ has infinitely many solutions.
 - C) There exists an invertible matrix E such that $EA = I$.
 - D) The determinant of A is zero.
- Q6.** For an over-determined system of linear equations $\mathbf{X}\mathbf{w} = \mathbf{y}$ where $\mathbf{X} \in \mathbb{R}^{m \times d}$ ($m > d$) and $\mathbf{X}^\top \mathbf{X}$ is invertible, the left-inverse of $\mathbf{X}$ is given by:
- A) $\mathbf{X}^\top (\mathbf{X}\mathbf{X}^\top)^{-1}$
 - B) $(\mathbf{X}\mathbf{X}^\top)^{-1} \mathbf{X}^\top$
 - C) $\mathbf{X}^\top (\mathbf{X}^\top \mathbf{X})^{-1}$
 - D) $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$
- Q7.** For an invertible matrix $\mathbf{A} \in \mathbb{R}^{d \times d}$, its inverse is given by $\frac{1}{\det(\mathbf{A})} \text{adj}(\mathbf{A})$. The adjugate matrix $\text{adj}(\mathbf{A})$ is the transpose of:
- A) The identity matrix
 - B) The cofactor matrix $\mathbf{C}$
 - C) The left-inverse matrix
 - D) The minor matrix $\mathbf{M}$
- Q8.** Which of the following represents the right-inverse of a matrix $\mathbf{X}$ for an under-determined system (assuming $\mathbf{X}\mathbf{X}^\top$ is invertible)?
- A) $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$
 - B) $\mathbf{X}^\top (\mathbf{X}\mathbf{X}^\top)^{-1}$
 - C) $\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}$
 - D) $(\mathbf{X}\mathbf{X}^\top)^{-1} \mathbf{X}$
- Q9.** State Bayes' rule for continuous random variables X and Y to compute $p(x | y)$.
- Q10.** Give the definition of Entropy $H_P(\mathcal{X})$ for a discrete random variable over a finite alphabet $\mathcal{X}$.
- Q11.** What is the condition for a collection of d -vectors $\mathbf{x}_1, \dots, \mathbf{x}_m$ to be linearly independent?
- Q12.** Write down the multivariate Gaussian distribution $\mathcal{N}(\mathbf{x} | \boldsymbol{\mu}, \boldsymbol{\Sigma})$ for a D -dimensional vector $\mathbf{x}$.
- Q13.** Given $P(A) = 0.3$, $P(B | A) = 0.6$, and $P(B | \neg A) = 0.2$. Calculate marginal probability $P(B)$ and conditional probability $P(A | B)$.

- Q14.** Let $\mathcal{X} = \{0, 1\}$ be a binary source with $P(X = 0) = 0.8, P(X = 1) = 0.2$. Let Q be another distribution with $Q(X = 0) = 0.5, Q(X = 1) = 0.5$. Calculate the Entropy $H_P(\mathcal{X})$, Cross-entropy $H_{P,Q}(\mathcal{X})$, and KL divergence $D_{P,Q}(\mathcal{X})$. (You can leave logarithmic expressions base 2 as $\log_2(\cdot)$).
- Q15.** Let the joint PDF of X and Y be $p_{X,Y}(x, y) = c(x + y)$ for $0 < x < 1$ and $0 < y < 1$, and 0 otherwise. Find the constant c , and then compute the marginal PDF $p_X(x)$.
- Q16.** Given the matrix $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$, find its determinant, cofactor matrix, adjugate matrix, and inverse $\mathbf{A}^{-1}$. (Note: The formula for matrix inversion will be provided in the exam.)
- Q17.** Consider the over-determined system $\mathbf{X}\mathbf{w} = \mathbf{y}$ where $\mathbf{X} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$. Compute the left-inverse $\mathbf{X}^\dagger = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ and find the approximate solution for $\mathbf{w}$.
- Q18.** Consider the under-determined system $\mathbf{X}\mathbf{w} = \mathbf{y}$ where $\mathbf{X} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$. Compute the right-inverse $\mathbf{X}^\dagger = \mathbf{X}^\top (\mathbf{X} \mathbf{X}^\top)^{-1}$ and find one possible solution vector $\mathbf{w}$.
- Q19.** Derive the relationship between Cross-entropy and Kullback-Leibler divergence for a discrete space $\mathcal{X}$: $H_{P,Q}(\mathcal{X}) = H_P(\mathcal{X}) + D_{P,Q}(\mathcal{X})$.
- Q20.** Prove that the variance of a random variable X can be expressed as $\text{Var}(X) = E(X^2) - (E(X))^2$, starting from its definition $\text{Var}(X) = E((X - E(X))^2)$.

Solutions

- Q1. B.** Since A and B are independent, $P(A \cap B) = P(A)P(B) = 0.4 \times 0.5 = 0.2$. Then, $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.4 + 0.5 - 0.2 = 0.7$.
- Q2. B.** The variance of a binomial distribution $\text{Bin}(N, \mu)$ is defined as $N\mu(1 - \mu)$.
- Q3. C.** KL divergence is related to cross-entropy by the equation $H_{P,Q}(\mathcal{X}) = H_P(\mathcal{X}) + D_{P,Q}(\mathcal{X})$, which implies $D_{P,Q}(\mathcal{X}) = H_{P,Q}(\mathcal{X}) - H_P(\mathcal{X})$. It is not symmetric and is non-negative ($D_{P,Q} \geq 0$).
- Q4. C.** The CDF is defined as the integral of the PDF from $-\infty$ to x . Note that $p_X(x)$ can be greater than 1, and $P(X = x) = 0$ for continuous random variables.
- Q5. C.** An equivalent statement for A being nonsingular is that there exists an invertible matrix E such that $EA = I$. Columns must be linearly independent, and the determinant must be non-zero.
- Q6. D.** The left-inverse is calculated as $\mathbf{X}^\dagger = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$.
- Q7. B.** $\text{adj}(\mathbf{A})$ is the transpose of the cofactor matrix $\mathbf{C}$.
- Q8. B.** The right-inverse is calculated as $\mathbf{X}^\dagger = \mathbf{X}^\top (\mathbf{X} \mathbf{X}^\top)^{-1}$.
- Q9.** $p(x | y) = \frac{p_{X,Y}(x,y)}{p_Y(y)} = \frac{p_X(x)p(y|x)}{\int_{-\infty}^{\infty} p_X(x')p(y|x')dx'}$.
- Q10.** Entropy is the expected value of information: $H_P(\mathcal{X}) = -\sum_{x_k \in \mathcal{X}} p_k \cdot \log(p_k)$.
- Q11.** A collection of vectors is linearly independent if $\beta_1 \mathbf{x}_1 + \dots + \beta_m \mathbf{x}_m = 0$ holds *only* for $\beta_1 = \dots = \beta_m = 0$.
- Q12.** $\mathcal{N}(\mathbf{x} | \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{\frac{D}{2}} |\boldsymbol{\Sigma}|^{\frac{1}{2}}} \exp\left(-\frac{(\mathbf{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})}{2}\right)$.
- Q13.** First, apply the law of total probability: $P(B) = P(B | A)P(A) + P(B | \neg A)P(\neg A) = 0.6 \times 0.3 + 0.2 \times (1 - 0.3) = 0.18 + 0.14 = 0.32$. Then, by Bayes' rule: $P(A | B) = \frac{P(B|A)P(A)}{P(B)} = \frac{0.6 \times 0.3}{0.32} = \frac{0.18}{0.32} = 0.5625$.
- Q14.** We use $\log_2$ for bits: Entropy: $H_P(\mathcal{X}) = -0.8 \log_2(0.8) - 0.2 \log_2(0.2) \approx 0.7219$ bits. Cross-entropy: $H_{P,Q}(\mathcal{X}) = -0.8 \log_2(0.5) - 0.2 \log_2(0.5) = -1 \log_2(0.5) = 1$ bit. KL divergence: $D_{P,Q}(\mathcal{X}) = 0.8 \log_2\left(\frac{0.8}{0.5}\right) + 0.2 \log_2\left(\frac{0.2}{0.5}\right) = 0.8 \log_2(1.6) + 0.2 \log_2(0.4) \approx 0.278$ bits. (Alternatively, $D_{P,Q}(\mathcal{X}) = H_{P,Q}(\mathcal{X}) - H_P(\mathcal{X}) = 1 - 0.7219 = 0.2781$).
- Q15.** The integral of the joint PDF over the entire domain must equal 1: $\int_0^1 \int_0^1 c(x+y) dx dy = c \int_0^1 \left[\frac{1}{2}x^2 + xy\right]_{x=0}^1 dy = c \int_0^1 \left(\frac{1}{2} + y\right) dy = c \left[\frac{1}{2}y + \frac{1}{2}y^2\right]_0^1 = c(1) = 1 \Rightarrow c = 1$. The marginal PDF is $p_X(x) = \int_0^1 1 \cdot (x+y) dy = \left[xy + \frac{1}{2}y^2\right]_0^1 = x + 0.5$ for $0 < x < 1$.
- Q16.** Determinant $\det(\mathbf{A}) = (2)(3) - (1)(1) = 5$. Cofactor matrix $\mathbf{C} = \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}$. Adjugate matrix $\text{adj}(\mathbf{A}) = \mathbf{C}^\top = \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}$. Inverse $\mathbf{A}^{-1} = \frac{1}{5} \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 0.6 & -0.2 \\ -0.2 & 0.4 \end{bmatrix}$.

Q17. First, compute $\mathbf{X}^\top \mathbf{X} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. The inverse $(\mathbf{X}^\top \mathbf{X})^{-1} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$.

The left-inverse $\mathbf{X}^\dagger = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & 1 & -1 \\ -1 & 1 & 2 \end{bmatrix}$. The approximated solu-

tion $\mathbf{w} = \mathbf{X}^\dagger \mathbf{y} = \frac{1}{3} \begin{bmatrix} 2 & 1 & -1 \\ -1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2(1) + 1(2) - 1(2) \\ -1(1) + 1(2) + 2(2) \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 2/3 \\ 5/3 \end{bmatrix}$.

Q18. First, compute $\mathbf{X}\mathbf{X}^\top = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. The inverse $(\mathbf{X}\mathbf{X}^\top)^{-1} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$.

The right-inverse $\mathbf{X}^\dagger = \mathbf{X}^\top (\mathbf{X}\mathbf{X}^\top)^{-1} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ 1 & 1 \\ -1 & 2 \end{bmatrix}$. The solu-

tion vector $\mathbf{w} = \mathbf{X}^\dagger \mathbf{y} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ 1 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2(2) - 1(4) \\ 1(2) + 1(4) \\ -1(2) + 2(4) \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 0 \\ 6 \\ 6 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix}$.

Q19. We expand the definition of cross-entropy:

$$\begin{aligned} H_{P,Q}(\mathcal{X}) &= - \sum_{x \in \mathcal{X}} P(x) \log Q(x) \\ &= - \sum_{x \in \mathcal{X}} P(x) \log Q(x) - \sum_{x \in \mathcal{X}} P(x) \log P(x) + \sum_{x \in \mathcal{X}} P(x) \log P(x) \\ &= - \sum_{x \in \mathcal{X}} P(x) \log P(x) + \sum_{x \in \mathcal{X}} P(x) (\log P(x) - \log Q(x)) \\ &= H_P(\mathcal{X}) + \sum_{x \in \mathcal{X}} P(x) \log \frac{P(x)}{Q(x)} \\ &= H_P(\mathcal{X}) + D_{P,Q}(\mathcal{X}). \end{aligned}$$

Which completes the derivation.

Q20. We expand the definition of variance using the linearity of expectation:

$$\text{Var}(X) = E((X - E(X))^2) = E(X^2 - 2XE(X) + (E(X))^2)$$

Since $E(X)$ is a constant, we can apply linearity:

$$\begin{aligned} &= E(X^2) - 2E(X)E(X) + (E(X))^2 \\ &= E(X^2) - 2(E(X))^2 + (E(X))^2 \\ &= E(X^2) - (E(X))^2 \end{aligned}$$

Which concludes the proof.